

Reward component ablation — macro-average (Nominal + High Load + Cold Winter)

Const-only
(c, w=0)

1.1%

VR (exceedance)

1798 m³/d

score 0.901 | term 0%

Linear-only
(w, c=0)

9.9%

VR (exceedance)

1984 m³/d

score 0.911 | term 0%

MS-SAC
(w+c)

17.2%

VR (exceedance)

1879 m³/d

score 0.748 | term 11%

ST-SAC
(large w, c=0)

10.1%

VR (exceedance)

1972 m³/d

score 0.897 | term 0%